Fiscal Year:	FY 2025	**Task Last Updated:**	FY 03/14/2025
PI Name:	Bhattacharjee, Subrata Ph.D.		
Project Title:	Residence Time Driven Flame Spread: The Final Phase		
Division Name:	Physical Sciences		
Program/Discipline:			
Program/Discipline--Element/Subdiscipline:			
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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Zip Code:	92182-0001	**Congressional District:**	53
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	Physical Sciences Unsolicited
Start Date:	06/01/2021	**End Date:**	03/31/2026
No. of Post Docs:	0	**No. of PhD Degrees:**	0
No. of PhD Candidates:	0	**No. of Master' Degrees:**	2
No. of Master's Candidates:	2	**No. of Bachelor's Degrees:**	0
No. of Bachelor's Candidates:	1	**Monitoring Center:**	NASA GRC
Contact Monitor:	Stocker, Dennis P	**Contact Phone:**	216-433-2166
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Flight Program:			
Flight Assignment:	NOTE: End date changed to 03/31/2026 per D. Stocker/NASA-GRC (Ed., 9/30/25).		
Key Personnel Changes/Previous PI:			
COI Name (Institution):	Miller, Fletcher Ph.D. (San Diego State University) Paolini, Christopher Ph.D. (San Diego State University) Takahashi, Shuhei Ph.D. (Gifu University, Japan)		
Grant/Contract No.:	80NSSC21K1126		
Performance Goal No.:			
Performance Goal Text:			

Task Description:	The continuing work builds upon our ground and space based studies so far – theoretical, computational, and experimental – to investigate steady and unsteady flame propagation over solid fuels in a microgravity environment. One of the central hypotheses of the Residence Time Driven Flame Spread (RTDFS) project is that steady flame spread over solid fuels cannot be sustained in a microgravity environment, even at high oxygen environment, if the fuel is thicker than a critical thickness. At lower oxygen level, a similar situation arises when the opposing flow velocity is below a critical velocity. A simplified theory based on scale analysis has produced closed-form formulas in terms of known parameters for these critical quantities. A comprehensive computational model that includes gas and surface radiation, finite-rate chemistry for pyrolysis, and combustion support these results. Moreover, computations reveal that the asymmetry between the thermal and species fields introduced by radiative losses is responsible for a slowly moving flame to be surrounded by a vitiated atmosphere leading to extinguishment. Earlier space based experiments showed a slowing flame over thick poly(methyl methacrylate)/PMMA slabs consistent with our findings. Also, experiments in the Burning and Suppression of Solids-II (BASS-II) project, conducted in 2019, has produced a wealth of data at a relatively low oxygen level that are consistent with our prediction. However, no space based experiments have ever been performed to test the hypothesis that steady flame spread over fuels exceeding a certain thickness is unsustainable even at high oxygen level. The proposed space based experiments in the Solid Fuel Ignition and Extinction (SoFIE) project are intended to establish the critical fuel thickness, critical flow velocity, and experimentally establish the mechanism of flame extinguishment. The current extension proposal is to continue our ground based effort for the successful execution of the flight experiments and processing the data from the flight experiments to test the proposed hypotheses. Some of this work includes: (a) Refining the numerical model to create grid and domain independent simulation of opposed-flow flame spread in four configurations: microgravity, thermal, kinetic, and downward. (b) Parametric studies involving fuel thickness, oxygen level, pressure, opposing-flow velocity, and ambient pressure. (c) Improve the pyrolysis kinetics model. (d) Evaluate the effect of solid-phase conduction in flame spread. (e) Develop expressions for flame length in different regimes. (f) Theoretical work to establish a comprehensive set of non-dimensional numbers, supported by computation, to delineate different regimes of flame spread. (g) Further develop the MATLAB based Flame Image Analysis Tool (FIAT) application to be able to analyze videos from the SoFIE experiments. (h) Conduct experiments in the Flame Tunnel and Flame Stabilizer at San Diego State University (SDSU) to test our numerical model. (i) Modify the Ignition Apparatus at SDSU to conduct ignition experiments to complement ignition delay data from all SoFIE experiments to develop an ignition model for solid fuel in a microgravity environment. (j) Process the data obtained from the Space experiments to test the hypothesis of the original proposal. (k) Further develop the educational outreach web portal, www.themofluids.net, currently used by over 3000 educators. (l) Publish the research result in renowned combustion journals and conferences. Establishing a critical non-dimensional parameter that leads to self-extinguishing flame in a low-gravity environment has a direct implication to fire safety in spacecrafts. It can produce a critical thickness, a critical velocity, a critical pressure, or oxygen level that makes a thin combustible material inherently fire-safe. Validating the mechanism of such self-extinguishing flames – that a flame chokes itself due to the asymmetry of its species and thermal fields, established numerically – from the RTDFS experiments will be a significant scientific discovery. NOTE: Project continues as the last phase of "Residence Time Driven Flame Spread Over Solid Fuels--NNX15AG11G".
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	Our research has three components: We have built a novel experimental facility called Flame Tower at SDSU, where ground based experiments on flame spread are being conducted; Theoretical and computational work that explore the similarity and differences between the mechanisms flame spread in a zero gravity space environment and on Earth; Design and propose a space based experiment to establish extinction mechanism of flames. The data that we are acquiring in our ground based work will provide the research community with a comprehensive set of results for testing different theories of flame spread and solid fuel ignition in a normal gravity environment. The space-based experiments will test our hypothesis that a flame spread over a flat solid fuel is unstainable if the fuel thickness exceeds a critical thickness. This will have implications in designing fire safe habitat in space. The space based experiments have just been completed and we are in the process of collating the data. We will start processing the data as soon as our Matlab based software is ready. These spaced based data is the first set of data on flame spread over PMMA in a completely quiscent environment. It will inspire new theories and further work to improve our understanding of flame spread in a microgravity environment. Work is continuing to improve our thermodynamic portal, www.thermofluids.net, which is used by a huge number of students and educators around the world.
Task Progress:	We have just concluded our experiments aboard the International Space Station. A number of tests have been conducted, including: 1. Flame spread over poly(methyl methacrylate)/PMMA at 40% and 30% oxygen level to determine the effect of ambient oxygen concentration on flame spread rate in a quiescent microgravity environment. 2. Tests to evaluate the fuel thickness effect on flame spread rate in a quiescent microgravity environment. 3. Tests to evaluate the effect of ambient pressure on flame spread rate in a quiescent microgravity environment. 4. Tests to evaluate interactions between two spreading flames approaching each other in a quiescent microgravity environment. 5. Tests to evaluate the effect of flow velocity for opposed flow flame spread in the absence of gravity. 6. Tests to evaluate the effect of flow velocity for concurrent flow flame spread in the absence of gravity.
Bibliography Type:	Description: (Last Updated: 03/31/2026)

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Articles in Peer-reviewed Journals	Bhattacharjee S, Panzer F. "Lateral flame spread over thermally thin fuels: Theoretical prediction of flame spread rate and experimental validation." J. Phys: Conf. Ser. 2024 Nov 1;2885(1):012003. http://dx.doi.org/10.1088/1742-6596/2885/1/012003 , Nov-2024
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Dissertations and Theses	Aurhammer S. "Theoretical analysis and experimental investigation of flame propagation via thermally thin and thick PMMA samples: Validation of theoretical models and evaluation of thermophysical correlations." Master's Thesis, Universität der Bundeswehr in Munich, Germany and San Diego State University, May 2024. , May-2024
Dissertations and Theses	Zielke S. "Influences on flame spread by the geometry and orientation of a fuel." Master's Thesis, Universität der Bundeswehr in Munich, Germany and San Diego State University, May 2024. , May-2024